DISHA JAIN

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Software Developer with experience building backend services, analytics platforms, and AI-powered applications. Hands-on experience with **NestJS, TypeScript, ClickHouse, MongoDB, Redis, LLMs, and RAG**, with a focus on scalable backend systems, data-heavy APIs, and AI-assisted product features.

Education

International Institute of Information Technology (IIITH) — **Hyderabad, Telangana**
M.Tech in CSIS - CGPA: 7.52/10.00 — *July 2023 - May 2025*

Shri Vaishnav Vidyapeeth Vishwavidyalaya — **Indore, Madhya Pradesh**
B.Tech in CSE - CGPA: 8.5/10.00 — *July 2018 - May 2022*

Experience

Coschool.ai - ***Software Developer*** — **March 2025 - Present**
EdTech platform - AI-powered learning for K-12 schools (students, teachers, parents)

Role-Based Analytics Dashboards | *NestJS, ClickHouse, MongoDB, Materialized Views, AWS SES*

- Built **role-based analytics dashboards** for teachers using **NestJS, ClickHouse and MongoDB**, tracking homework completion and academic performance across classes, sections and subjects.
- Improved dashboard response times by **58%** by reducing expensive database work and using **ClickHouse materialized views** to pre-compute frequently used student activity metrics.
- Built **one-click dashboard actions** for principals and teachers to act on student insights directly, including **nudges for inactive students, login reminders, appreciation emails**, and assigning **level-up homework** to students marked **Needs Improvement**.

Student Weekly Progress Reporting System | *NestJS, MongoDB, ClickHouse, Redis*

- Built an automated **weekly progress report sent to parents of 90K+ students**, summarizing what their child learned during the week, including **class updates, learning time, homework, self-study, doubts cleared and experiential activities**.
- Generated **personalized and actionable insights for parents** based on each student's activity, including inactivity alerts, weak chapters, overdue homework, pending level-ups and learning-credit reminders.
- Automated report generation every week using **scheduled cron jobs**, storing each report as a **fixed historical snapshot** and using **Redis caching** for fast access to previously generated reports.

Agentic Question Generation & Validation | *LLMs, Prompt Engineering, AWS S3, Python, LangGraph*

- Built a **Question generation agent using LangGraph** from teacher-uploaded chapter PDFs to generate **chapter aligned questions** from the extracted content.
- Designed a **Question validation agent** for generated and teacher-uploaded **questions**, checking **correctness, completeness and chapter/topic relevance** before publication.
- Generated **context-aware concept explanations** using question content and curriculum metadata, keeping explanations aligned with the selected chapter and topic.
- Automated **step-by-step solution generation** and **rubric-based validation** to improve the consistency and quality of student-facing solutions.

Practice & Mock Test Engine | *NestJS, TypeScript, ClickHouse, MongoDB*

- Built the **end-to-end Practice and Mock Test engine**, allowing students to **pause and resume tests, continue across devices and retain saved answers**, while handling test-specific time limits and inactivity timeouts.
- Ensured a student can have only **one active test session at a time**, preventing duplicate attempts and conflicting answers when the same test is opened from another device or tab.
- Built detailed **post-test reports** showing score, accuracy, time spent per question, revisits and answer changes to help students understand where they lost marks.

Personal Projects

P2P File Sharing System | *C++, Socket Programming, Multi-threading*

- Built a **group-based peer-to-peer file sharing system** with socket communication, user authentication, **SHA-1 integrity checks**, and multi-threaded simultaneous downloads.

Scalable Media Conversion | *Python, RabbitMQ, MongoDB*

- Developed a **microservices-based video-to-audio conversion platform** with JWT authentication, asynchronous RabbitMQ messaging, load balancing, and email alerts.

Technical Skills

Languages:	TypeScript, JavaScript, HTML, CSS, Python, C++, C
Databases:	ClickHouse, MongoDB, MySQL, PostgreSQL, Redis, DynamoDB
Frameworks/Tools:	NestJS, Node.js, Express.js, Kafka, AWS, Locust
AI:	LLMs, RAG, Prompt Engineering
Coursework:	Operating Systems, Computer Networks, Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming

Achievements

- Secured **96.79 percentile** in GATE CS 2023.
- Ranked in the **top 100** in PGEE 2023.
- Qualified **BARC OCES/DGFS 2023**.
- **IIITH Dean's List**, Monsoon 2024.